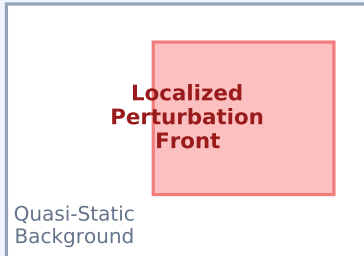


Stateful Selective Maintenance of Two-Level Schwarz Preconditioners

Persistent Factor Reuse and Coarse Tracking for Evolving Sparse Linear Systems ($A_t x_t = b_t$)

1. Evolving Operator

Transient Continuum PDEs



$$A_{t-1} \rightarrow A_t$$

- Spatially localized drift
- Moving phase / damage
- Ambient factors stale
- Full rebuild wasteful
- Blind freeze diverges

2. Online Monitoring

Inexpensive Drift Sensing

$$d_i^{\text{diag}} = |\text{diag}(\Delta A_i)|_2$$

Memory scan: $O(n_i)$
vs. $O(\text{nnz}_i)$ Frobenius

Overhead $\leq 0.28\%$

- Exact Pearson $r = 1.00$
- Spearman $\rho = 0.79-0.98$
- Strong rank agreement
- Elapsed time < 0.22 ms
- Negligible sensing tax

3. Stateful Repair

Persistent Object Policy

mass-alpha Truncation

Refactor Disturbed S_t
Reuse Stable Factors

Galerkin Coarse: $A_0(t)$

- PETSc/MUMPS symbol reuse
- Numeric refactor: 25%-32%
- Zero rebuild for 68%-75%
 - Global error controlled
 - Age staleness penalty

4. Certified Gain

Flagship 48^3 (110k DOFs)

-63.9% Setup

19.6%-21.8% Net Saved
1.24x ~ 1.28x Speedup

Krylov: 50.7 vs 51.0

- Zero iteration penalty
- RelRes $\leq 3.89e-10$
- Pass strict ($< 1e-8$)
- Scales as $n_{\text{sub}}^{1.43}$
- Release: v1.2-cmame-final